SOC ANALYST LEVEL 1

Case Management & SOC Workflow

Professional Handling of Alerts, Tickets, and Escalations

Nader Anam | 3 Hour Lecture

LECTURE OBJECTIVES

- Define Case Management within the SOC.
- Contrast Alert, Ticket, Case, and Incident.
- Understand the standard Case Lifecycle.
- Learn documentation standards for Notes.
- Manage Ticket statuses efficiently.
- Apply SLA metrics and Prioritization.
- Communicate with Tier 2, IT, and Users.
- Execute professional Shift Handovers.
- Manage high-pressure alert queues.
- Execute documented Case Closures.

IMPORTANCE OF WORKFLOW

Technical analysis is only half the work. The rest is operational management.

Input

Alert arrives and a Ticket is formally opened.

Process

Evidence is gathered and professional Notes are logged.

Output

Escalation or Closure is finalized with a report.

COMMON BEGINNER PITFALLS

Documentation Errors

Analyzing an alert but leaving the notes section empty.

Closing a case without explaining "Why."

Workflow Errors

Escalating to Tier 2 without providing evidence.

Ignoring SLA timers during shift changes.

DEFINING CASE MANAGEMENT

Managing a security situation from the moment it triggers until the final cleanup.

1
Intake

Opening, classifying, and prioritizing the alert.

2
Triage

Evidence gathering, note taking, and updating status.

3
Resolution

Escalation, final closure, and archiving evidence.

COMPARING CORE TERMS

Term	Technical Context	System Example
Alert	Signal from SIEM/EDR.	"Suspicious PowerShell"
Ticket	A record of work/time.	"ServiceNow #INC123"
Case	Broad investigation.	"Host Analysis for PC-01"
Incident	Confirmed breach.	"Ransomware Execution"

FOCUS ON: ALERTS

Alerts are the noise that requires human ears. They are unverified triggers.

- Failed Login attempts.
- Malware signatures detected.
- Large data transfers.
- Audit logs being cleared.
- C2 communication patterns.
- Impossible travel logins.

FOCUS ON: TICKETS

The official tracking log. If it isn't in the ticket, it didn't happen.

Metadata

ID, Date, Time, Analyst name.

Status

New, In-Progress, Escalated.

Audit

Every action and comment history.

FOCUS ON: CASES

A Case links related alerts together. It maps the attack chain.

The Phishing Campaign Example

1 Phishing Email → 50 recipients → 5 link clicks → 1 malware alert.

Manage all 57 separate events inside **ONE investigation Case**.

FOCUS ON: INCIDENTS

An Incident is a high-priority situation requiring a Response Team.

- Compromised Admin account.
- Successful data exfiltration.
- Ransomware active spread.
- Persistence established on DC.
- Server shell access confirmed.
- Credential theft confirmed.

THE CASE LIFECYCLE

Stage 1: Beginning

- New: Alert arrives in queue.
- Assigned: Ownership established.

Stage 2: Investigation

- In Progress: Active work.
- Pending Info: Waiting for others.

Stage 3: Decision

- Escalated: Higher Tier review.
- Resolved: Threat neutralized.

Stage 4: Archiving

- Closed: Final record.
- Reopened: New evidence found.

STATUS: NEW

The queue entry stage. Do not ignore "New" alerts.

- Check the source: Is it a sensitive server?
- Check the Severity: High/Critical gets eyes first.
- Check the time: How long has it sat in the queue?
- **Action:** Accept the case or assign to an available peer.

STATUS: ASSIGNED

Ownership is everything. One case, one owner.

The Owner's Duty:

Review the full log history immediately. Formulate a hypothesis. Begin the investigation. **Update the ticket status to 'In Progress' within 5 minutes.**

STATUS: IN PROGRESS

The investigative deep-dive phase.

Queries

Executing KQL/SPL to find related events.

Timeline

Mapping what happened and when.

Notes

Logging every finding as it happens.

STATUS: PENDING INFORMATION

You are waiting for something outside the SOC's direct control.

- Waiting for a user to confirm a login attempt.
- Waiting for IT to verify a scheduled maintenance task.
- Waiting for Tier 2 to finish malware analysis.

Always set a follow-up reminder to avoid "Zombie Tickets."

STATUS: ESCALATED

Handing off the case to specialists or higher-ranking analysts.

The Hand-off Package:

1. Concise Summary. | 2. Key Evidence (IOCs). | 3. Initial Timeline. | 4. Reason for Escalation. | 5. Recommended Actions.

STATUS: RESOLVED

The work is done, the threat is neutralized, but the file is not archived yet.

- ✓ Verify all containment steps were successful.
- ✓ Check that the malicious activity has not returned.
- ✓ Ensure the user is back online and secure.
- ✓ Draft the final remediation report.

STATUS: CLOSED

The permanent historical record. Every closure needs a Disposition.

Verdict

FP, Benign, or TP?

Reason

Why was it closed?

Evidence

Final list of IOCs.

STATUS: REOPENED

Sometimes cases come back from the dead.

> A case is reopened if the same user triggers the same alert within 24 hours, or if Tier 2 finds a hidden backdoor you missed in your initial investigation.

WHAT IS AN SLA?

Service Level Agreement: The contract of speed.

Priority	ACK Time	Resolution Time
Critical	15 Min	4 Hours
High	30 Min	8 Hours
Medium	2 Hours	24 Hours
Low	1 Day	3 Days

SLA IMPORTANCE

- Ensures consistent response quality.
- Prioritizes critical server alerts.
- Measures SOC team performance.
- Triggers auto-escalation to managers.

PRIORITIZATION FACTORS

Context determines Priority, not just the name of the alert.

Asset Value

DC vs Contractor Laptop.

Privilege

Admin vs Guest User.

Scope

1 device vs 50 devices.

PRIORITIZATION EXAMPLE

Alert 1: Single Failed Login for standard user.

Priority: Low

Alert 2: Audit Logs Cleared on Domain Controller.

Priority: CRITICAL

THE INCIDENT TIMELINE

A chronological sequence of the attack life.

```
08:00 - Phishing Email Delivered
08:15 - User Clicked Link (Proxy Log)
08:16 - Credentials Submitted (Identity Log)
08:30 - Successful RDP from External IP (Windows Log)
08:45 - Malicious Rule Created (Exchange Log)
```

PROFESSIONAL CASE NOTES

Notes are the analyst's brain on paper. They must be:

- ✓ Objective (Facts only).
- ✓ Concise (No fluff).
- ✓ Structured (Checklist style).
- ✓ Chronological.

BAD NOTES EXAMPLE

> "Investigated the alert for finance.manager. It looks a bit weird. User says they didn't do it. Escalating to the next team to see if they can fix it."

Missing: Source IP, Time window, specific Event IDs, Next steps.

GOOD NOTES EXAMPLE

"Reviewed auth logs for finance.manager. Observed 18 failed logins from IP 185.20.40.9 followed by 1 successful RDP login. Event ID 4672 assigned. User confirmed not traveling. Escalating to Tier 2 for compromise response."

CASE NOTE TEMPLATE

```
[Time] - [Analyst Name]
Scope: [What was reviewed?]
Findings: [What was found? IPs/Hashes/IDs]
Verdict: [Initial Decision]
Next Step: [What happens now?]
```

TICKET UPDATE LOGIC

Never leave a ticket un-updated for more than 4 hours.

- ✓ Update when evidence is found.
- ✓ Update when a user is contacted.
- ✓ Update when you change the priority.
- ✓ Update before you go on break or end shift.

SOC COMMUNICATION

The SOC analyst is a hub. You speak to:

Tier 2

Technical escalation and malware analysis.

IT Support

Host isolation and password resets.

End Users

Confirmation and awareness training.

TIER 2 COMMUNICATION

Do not send Tier 2 a "Problem." Send them a "Case."

- State the exact reason for escalation.
- Link the specific logs you found in the SIEM.
- Specify what you need (e.g. Memory Forensics).

TIER 2 MESSAGE EXAMPLE

```
"Escalating SOC-1045: Suspected Account Compromise. User HR_Nader. 15 failed logins → 1 success from anomalous IP 1.2.3.4. Excel later spawned encoded PowerShell. Requested Tier 2 review of endpoint memory and network persistence."
```

COMMUNICATING WITH IT

IT teams are busy. Be direct and instructional.

> **Instruction:** "Please isolate host FIN-PC-01 from the production VLAN immediately due to active Malware execution. Ticket INC-500 ref."

END USER ETIQUETTE

Professionalism over Panic.

- **DO:** "We are verifying account security. Did you log in from Paris today?"
- **DON'T:** "Your PC is full of viruses and you are a liability to the network."

CRUCIAL USER QUESTIONS

- Did you open an email titled [X]?
- Did you enter your password?
- Were you traveling?
- Did you open the attachment?
- Did you click 'Enable Macros'?
- Did you notice any strange pop-ups?

USER COMMUNICATION: DONTS

- Never ask for a user's password.
- Never blame the user for a breach.
- Never share confidential attacker infrastructure IPs with the user.
- Never tell a user "It's 100% fine" if you aren't sure.

CASE ASSIGNMENT LOGIC

Cases are assigned to ensure no alert goes un-owned.

Load Balance

Ensure no analyst is overwhelmed.

Skill Match

Assign phishing to the Phish specialist.

Availability

Ensure the owner is on shift.

OWNERSHIP BOUNDARIES

You own the case from Triage to Escalation/Closure.

> If you escalate a case to IT, you still own the ticket. You must follow up with IT until the host is clean and the case is closed. **Don't Escalate and Forget.**

THE SHIFT HANDOVER

Critical for 24/7 SOC operations. Transfer the "Current State."

- ✓ Current open cases count.
- ✓ Ongoing high-severity incidents.
- ✓ SLA deadlines approaching.
- ✓ Pending IT or user replies.

HANDOVER TEMPLATE

```
Case ID: [Number]
Priority: [High/Crit]
Summary: [What happened]
Blocker: [Why is it still open?]
Next Action: [What should the next guy do?]
```

HANDOVER EXAMPLE

> "SOC-2026-1045 (High). Suspected Admin Compromise. Successful RDP after brute force. User contacted, awaiting reply. Next Analyst: Monitor 185.x.x.x for new session activity."

TYPES OF ESCALATION

Functional

Escalating to a team with different permissions (Network/Cloud).

Hierarchical

Escalating to a Manager because a major incident is ongoing.

FUNCTIONAL ESCALATION

When you need the "Keys" to another kingdom.

- IAM Team: To reset a password.
- Network Team: To block a firewall port.
- Email Team: To delete a message from all mailboxes.

HIERARCHICAL ESCALATION

Bringing in the "Big Guns" or Management.

- When an SLA is about to breach.
- When several critical servers are down.
- When there is a massive data leak suspected.

CASE CLOSURE CHECKLIST

- ✓ All IOCs blocked/recorded?
- ✓ User account secured?
- ✓ Endpoint isolated/cleaned?
- ✓ SIEM rules tuned to prevent future FPs?
- ✓ Executive Summary written?

STANDARD DISPOSITIONS

False Positive

Alert triggered on normal activity.

Benign

Real activity but authorized (e.g. PenTest).

PROFESSIONAL CLOSURE NOTE

```
"Closed as Benign True Positive. Observed port scanning from 10.0.0.5. Confirmed with IT as an approved internal vulnerability scan during the maintenance window. No further follow-up required."
```

HANDLING DUPLICATE CASES

Do not just delete duplicates.

- Identify the Parent Case ID.
- Port any unique evidence (e.g. a new Source IP) to the Parent.
- Close the duplicate as "Duplicate of Case [Parent ID]."

CASE QUALITY METRICS

A good analyst is measured by the quality of their tickets.

- Accuracy of categorization.
- Clarity of investigation steps.
- Presence of specific IOCs.
- Adherence to SLA timers.

SOC PERFORMANCE METRICS

Alert Count

Volume of triggers by source.

Resolution Time

Average time to close cases.

FP Rate

Percentage of noisy false alarms.

UNDERSTANDING MTTD & MTTR

MTTD

Mean Time To Detect: How fast we catch the trigger.

MTTR

Mean Time To Resolve: How fast we kill the threat.

MANAGING THE BACKLOG

A backlog is an analyst's nightmare.

- Stop and Prioritize: Handle Criticals first.
- Batch triage: Group similar alerts (e.g. all 4625s).
- Escalate tuning requests to the SIEM engineers for noisy rules.

ALERT PRESSURE HANDLING

When the dashboard is red:

Focus

Ignore Lows until Highs are triaged.

Briefing

Update the Lead on current queue size.

WHAT IS A CAMPAIGN CASE?

A **Campaign** is a series of related malicious actions across the network using the same infrastructure (IP/Domain/Payload). Instead of 100 tickets, manage 1 Campaign Case with 100 related entities.

CAMPAIGN EXAMPLE

Case: *"2026 Q1 Salary Update Phishing Campaign"*

- ✓ Centralized IP Blocking.
- ✓ Centralized Inbox Purging.
- ✓ Mass User Notification.
- ✓ Consolidated IR Report.

EXERCISE 1: CASE NOTE

Alert: Login from New Country (Germany) for user Sara. MFA Passed. User confirms they are not in Germany.

Requirement: Write a professional Case Note.

ANSWER 1: PROFESSIONAL NOTE

```
"Reviewed auth logs for user Sara. Login originated from Germany. User confirmed she is NOT traveling. Suspicion of MFA bypass or token theft. Severity set to HIGH. Escalating to Tier 2 for session revocation and account review."
```

EXERCISE 2: ESCALATION

User Ahmed. 15 Failed logins → Success. PowerShell executed. C2 connection active to external IP.

Requirement: Write a Tier 2 Escalation message.

ANSWER 2: ESCALATION

```
"URGENT: Escalating confirmed compromise for User Ahmed. Evidence shows brute force success followed by encoded PowerShell execution and beaconing to 45.77.20.10. Initial containment recommended. Please review for persistence."
```

EXERCISE 3: CLOSURE

Internal Port Scan from 10.0.0.55. Confirmed it is the authorized Security Scanner during its maintenance window.

Requirement: Write a Closure Note.

ANSWER 3: CLOSURE

```
"Closed as Benign True Positive. Port scan activity mapped to authorized scanner 10.0.0.55. Verified scan schedule in IT change log. No suspicious activity followed."
```

EXERCISE 4: HANDOVER

Case ID: SOC-2026-2050. Status: Escalated. Priority: High. Awaiting Tier 2 review of endpoint logs.

Requirement: Write a concise Handover entry.

ANSWER 4: HANDOVER

```
"Case SOC-2050 (High) remains open. Escalated to Tier 2 for host forensics. Pending analysis of encoded PS script. Next analyst: Follow up with Tier 2 and check User Sara's status."
```

MASTER TRAINING SCENARIO

1. Alert: Single Failed Login.
2. Alert: Audit Logs Cleared on DC-01.
3. Alert: Malware Blocked (no execution).
4. Alert: Phishing Campaign (100 users, 2 credential thefts).

Order these by Priority and determine which need a Case immediately.

ANSWER: SCENARIO PRIORITY

#1: Audit Logs Cleared (DC-01) - Infrastructure threat.

#2: Phishing Campaign (Credential theft) - Identity threat.

#3: Malware Blocked - Contained host threat.

#4: Single Failed Login - Low noise.

ANSWER: CASE NEEDS

All items except #4 (Single Failure) require a formal Case.

- DC Audit Clear: Requires an Incident/Case due to Criticality.
- Phishing: Requires a Campaign Case to track the 100 targets.
- Malware: Requires a host Case to verify cleanup.

ANSWER: ESCALATIONS

Escalate to Tier 2

Audit Log Clear and Credential Theft.

Escalate to IT

Malware host isolation.

ANSWER: THE CRITICAL NOTE

```
"CRITICAL Alert on DC-01. Security Audit Logs cleared at 11:30 UTC. No approved change request found. Potential defense evasion by threat actor. Escalating to Tier 2 / Incident Response immediately."
```

ANSWER: MASTER HANDOVER

```
"Shift state: Critical DC-01 incident in progress (Escalated). High Phishing Campaign in progress (IAM contacted). Malware on SALES-PC contained. Queue size: 12 New alerts triaged."
```

COMMON WORKFLOW ERRORS

- Closing cases with 1-word notes.
- Missing SLA timers.
- Assigning cases to people on leave.
- Ignoring the 'Pending' queue.
- Failing to document user interactions.
- Escalating without a summary.

THE SOC GOLDEN RULE

"Perfect technical analysis is invisible if the case documentation is incomplete."

Every case must have an Owner, Notes, a Decision, and a Next Step.

LECTURE SUMMARY

- ✓ Alert → Ticket → Case progression.
- ✓ SLA and Priority context.
- ✓ Pillars of Case Documentation.
- ✓ Internal and External Communication.
- ✓ Shift Handover excellence.
- ✓ Logical Case Closure.

REVIEW QUESTIONS

1. What is the difference between a Ticket and a Case?
2. Why is a 'Pending Info' status dangerous if not monitored?
3. What are the 4 requirements of a good Tier 2 escalation?
4. Why is a Shift Handover vital in a 24/7 SOC?

MINI QUIZ

1. SLA stands for:
A) Secure Login Area | B) Service Level Agreement | C) System Log Alert

2. If a case is a Duplicate, you should:
A) Delete it. | B) Close it and link to the Parent. | C) Ignore it.

Answers: 1-B, 2-B

HOMEWORK ASSIGNMENT

You have 6 alerts in the queue (Phishing Campaign, Malware Blocked, HR Manager login failure...).

> Task: Order them by priority. Write a Case Note for the highest priority. Draft a Shift Handover for the remaining open tickets.

HOMEWORK TEMPLATE

```
Priority Order: [1-6]
Case Note (Top Priority): [Body Text]
Handover Entry: [Body Text]
```

MODEL HW KEY: PRIORITY

- Audit Log Clear (DC-01)
- HR Manager login denial.
- Phishing (1 cred theft).
- Port Scan (Unknown).
- Malware Blocked.
- Single Failure.

MODEL HW KEY: CASE NOTE

```
"High Alert: Login from New Country for HR Manager. User contacted via phone; confirmed they are home and did NOT log in. Potential compromise. Password reset initiated. Tier 2 notified."
```

MODEL HW KEY: ESCALATION

Audit Log Clear → Tier 2 Escalation message sent with DC-01 timestamp and pre-clearance events listed.

MODEL HW KEY: CLOSURE

Malware Blocked on USER-01 closed as "True Positive - Blocked." Verified no execution or network beaconing.

MODEL HW KEY: HANDOVER

```
"DC-01 investigation remains open with Tier 2. HR Manager password reset complete, awaiting session audit. Port Scan analysis pending source confirmation. All other alerts triaged."
```

Next Lecture:

SOC Tools & Daily Ops

A Day in the Life: EDR, SIEM, and Threat Intel management.